The safety filter (v9, now removed)

What it did, why it only reduced acceleration, and what it changed

1. The idea in one paragraph

The DIPF potential field decides the ego's acceleration. The safety filter was a small layer **after** it: every 0.1 s it checked the vehicles around the ego and asked “if I apply what DIPF wants, can I still always stop in time?”. If yes, it changed nothing. If no, it changed the command **as little as possible** to make it safe again. It used no lanes, only the bodies of the vehicles (the traffic is lane-free, with bikes filtering between cars), and it took each vehicle's type into account.

Order of operations in one control step:

- DIPF computes the acceleration from the potential field ($-\nabla U$).
- The acceleration limits are applied: forward ≤ 3.2 m/s², braking ≤ 4.5 m/s², sideways ≤ 3.0 m/s².
- **The safety filter** checks the surroundings and, if needed, lowers the forward acceleration (possibly into braking, never beyond 4.5 m/s²) and bounds the sideways acceleration.
- The speed limits are applied: 34 m/s forward, 3 m/s sideways, heading ≤ 15°.

2. The forward rule (keeping a safe distance)

Who is watched. Every vehicle ahead whose body overlaps the ego's path, or will overlap it within the next 1.5 s if both keep their sideways speed. A bike partly in the path counts; a truck in the next “lane” does not.

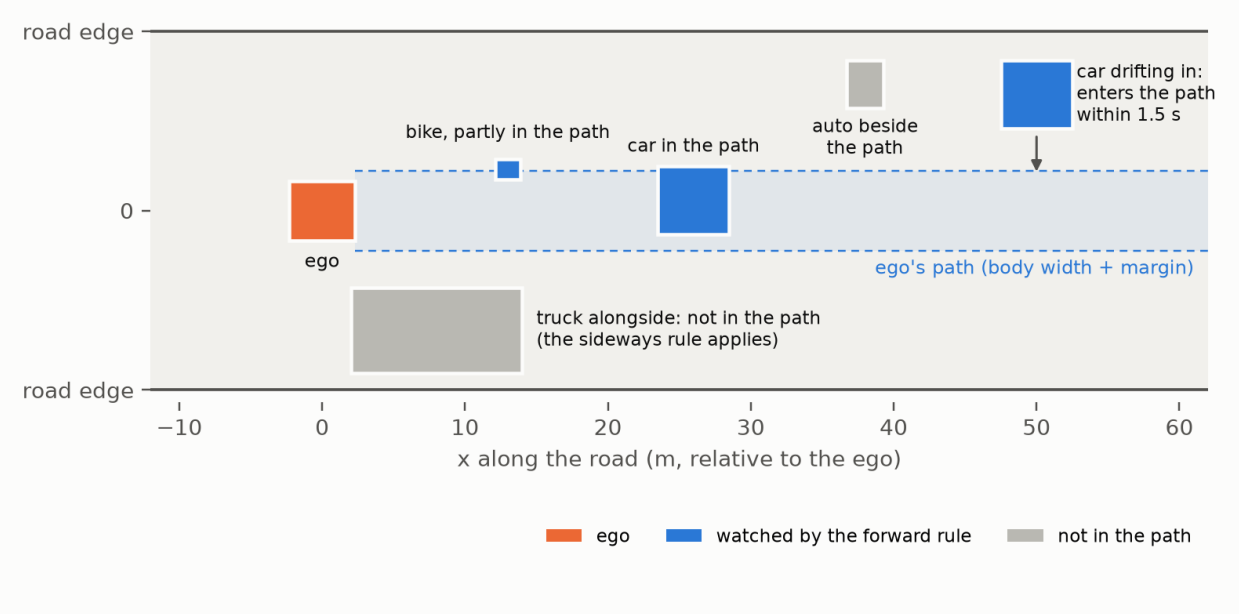


Figure 1. Which vehicles the forward rule watches. No lanes are used: a vehicle counts if its body overlaps the ego's path now, or is predicted to within 1.5 s.

The safe distance to each watched vehicle is the gap the ego needs so that, even if that vehicle brakes as hard as it can, the ego can still brake and stop behind it:

$$d_{safe} = d_0 + v_{ego} \cdot T_h + \max(0,\ v_{ego}^2 / (2\, b_{ego}) - v_{lead}^2 / (2\, b_{lead}))$$

d_0 = standstill gap (2 m) · T_h = time headway (1 s) · the last term = how much longer the ego needs to stop than the vehicle ahead

The vehicle ahead's braking ability b_{lead} depends on its **type** (from the vehicle definitions); the ego brakes at most 4.5 m/s²:

vehicle	bike	auto	car	bus	truck	ego
braking (m/s²)	7.0	5.0	4.5	4.0	4.0	4.5

So the ego keeps more distance behind a bike (which can stop quickly) than behind a bus.

The rule. Call the spare distance h = gap − d_{safe}. As long as $h \geq 0$ the ego can always stop in time. The filter lets h shrink, but only gradually, so it approaches zero ever more slowly and never crosses it (a *control barrier function*):

$$dh/dt \geq -\gamma \cdot h \quad (\gamma = 1 \text{ per second})$$

Writing out how h changes over time gives:

$$dh/dt = -(\text{closing speed}) - (T_h + v_{ego}/b_{ego}) \cdot a_{ego}$$

The number in front of the ego's acceleration a_{ego} is **negative**: accelerating more always makes the spare distance shrink faster. So the rule turns into an **upper limit** on the acceleration:

$$a_{ego} \leq a_{cap} = (\gamma \cdot h - \text{closing speed}) \,/\, (T_h + v_{ego}/b_{ego})$$

Why it only reduces acceleration. The acceleration closest to what DIPF asked for that respects an upper limit is simply **min(DIPF's request, a_{cap})**. If DIPF asks for less than a_{cap}, nothing changes; if it asks for more, it is cut to a_{cap}. Accelerating *more* never makes the ego safer with respect to a vehicle ahead, so there is never a reason to add acceleration.

Worked example. Ego at 14 m/s, a car ahead at 10 m/s, DIPF asking for +3.16 m/s². Safe distance = 2 + 14 + 10.7 = 26.7 m:

gap	spare distance h	a_{cap}	filter output	meaning
60 m	+33.3 m	+7.1 m/s²	+3.16 m/s²	cap above the request: no change
40 m	+13.3 m	+2.3 m/s²	+2.3 m/s²	still speeds up, just less
20 m	−6.7 m	−2.6 m/s²	−2.6 m/s²	too close: brakes until h is back to 0

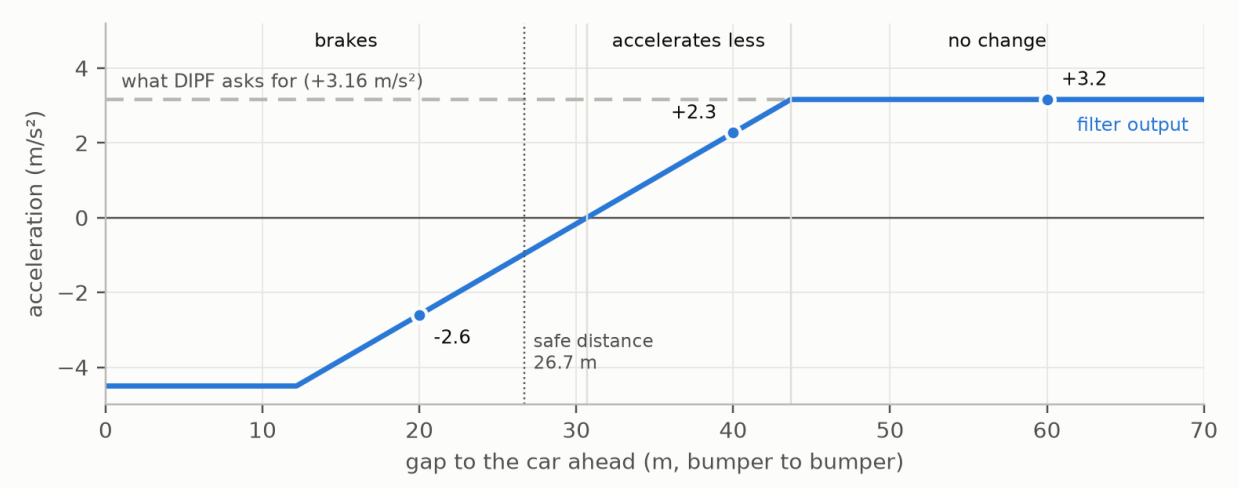


Figure 2. Filter output for the example above, as a function of the gap. Far away it passes DIPF's request through unchanged; closer in it trims the acceleration; inside the safe distance (and slightly beyond, while still closing) it brakes, never harder than 4.5 m/s².

3. The sideways rule

For vehicles **alongside** the ego (or about to be) and for the **road edges**, the ego may only move towards them as fast as it could still stop its sideways motion before a small margin:

sideways approach speed ≤ $\sqrt{(\ 2 \cdot a_{y,max} \cdot (\text{gap} - \text{margin})\ )}$ ($a_{y,max}$ = 3 m/s², margin = 0.3 m)

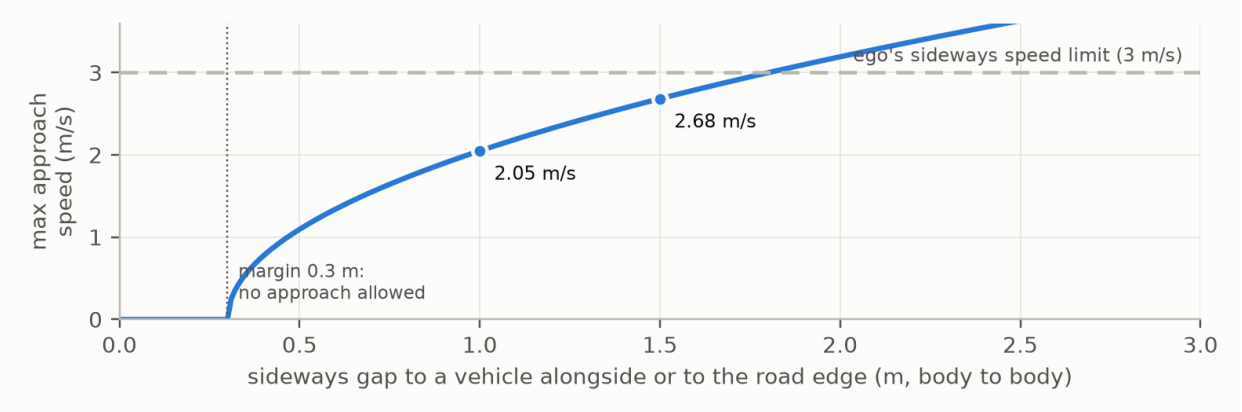


Figure 3. The fastest allowed sideways approach for a given sideways gap. Inside the margin no approach is allowed at all.

Unlike the forward rule, this one can come from **both sides** (a vehicle on the left gives an upper limit, one on the right a lower limit), so the filter keeps the sideways acceleration between the two. If DIPF pushes the ego towards a neighbour too fast, the limit can even **reverse** the push. If the ego is squeezed from both sides, it takes the middle.

4. Its settings

setting	value	meaning
sf_headway_s (T_h)	1.0 s	time headway in the safe distance; tuned by the optimiser
sf_min_gap_m (d_0)	2.0 m	gap kept to a stopped vehicle in the path; tuned
sf_lat_margin_m	0.3 m	sideways clearance kept to vehicles and road edges; tuned
gamma (γ)	1.0 /s	how fast the spare distance may shrink
horizon_s	1.5 s	how far ahead sideways motion is predicted for “in the path”
lead_brake	by type	braking assumed for the vehicle ahead (table in section 2)
override_jerk	on	applied after the jerk limit, so emergency braking is not delayed

5. What it changed

On 200 fresh seeds (1000–1199), full 1000 m road, before any re-tuning:

controller	success	collisions
v6 parameters + **safety filter**	**62.5%**	37.5%
same parameters, filter off	16.5%	83.5%
old v6 controller	13.0%	87.0%

Paired over the same seeds, the filter turned 99 failures into successes and 7 successes into failures (exact McNemar test $p < 0.0001$).

What the optimiser then learned. The DE search (before it was stopped) found that the best sets push towards the goal at almost full forward acceleration (sx = 10, i.e. +3.16 m/s²), switch the obstacle potential practically off (λ_{obs} = 0.01), and let the filter adjust the acceleration in about 94% of steps, mostly trimming it to hold the safe distance, like adaptive cruise control. Training success reached 98.3% on 60 seeds (no validation result: the run was deleted). In other words, the filter, not the potential field, was doing most of the driving.

What it did not cover:

- a vehicle ahead braking harder than its type's normal limit (SUMO allows emergency braking up to 9 m/s²; setting lead_brake to “emergency” would assume that);
- being hit from behind: it only controls the ego, so it cannot stop a follower from running into it;
- it is a guarantee only under its assumptions (constant sideways motion over 1.5 s, the braking values above).

6. Status

Removed from v9 and v10 on request, so both are pure DIPF again (about 16% success with the starting parameters). It can be re-added as a switch (off by default) if a comparison is needed.